

Mitigation of NH₃ emission from pig house using a rice husk biochar column and fertilizer potential of collected biochar

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Highlights

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Rice husk biochar (RHB) has NH₃ adsorption capacity.

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H₂SO₄-mixed RHB adsorbs more NH₃.

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Columns packed with RHB efficiently reduce NH₃ emission from pig house.

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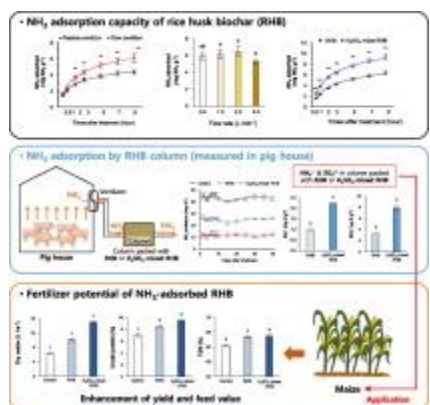
NH₃-adsorbed RHB collected from column can be recycled to support maize growth.

Abstract

NH₃ emissions have significant implications for ecosystems and human health. Pig houses are important sources of NH₃ emissions. This study aimed to evaluate the NH₃ adsorption capacity of rice husk biochar (RHB), its effectiveness in mitigating NH₃ emission from pig house using a RHB column, and the fertilizer potential of the collected RHB for maize growth. In a laboratory assay, NH₃ adsorption by RHB was measured using a closed chamber where NH₃ was diffused from NH₄OH. NH₃ adsorption in RHB packed in column was the highest at a flow rate of 2.0 L min⁻¹ (6.52 mg g⁻¹). H₂SO₄-mixed RHB enhanced NH₃ adsorption by 41.7% compared to RHB alone. When compared the efficiency of RHB column in NH₃ emission, the emission reduced by 34.5% and 58.2%, respectively, after passing RHB and H₂SO₄-mixed RHB columns compared to that measured without column (0.59 g d⁻¹). In a pig house setting, NH₃ concentration after passing the columns of control,

RHB, and H₂SO₄-mixed RHB was in average 14.8, 8.5, and 3.9 mg m⁻³, respectively, during experiment period. Cumulative emissions were reduced by 42.6% and 73.3% using RHB and H₂SO₄-mixed RHB columns, respectively. The application of RHBs collected from the column significantly increased dry matter and crude protein production in maize, with notable enhancement by H₂SO₄-mixed RHB application. These results demonstrate the integrated effectiveness of RHB application in NH₃ management, in which NH₃ emissions from the pig house are adsorbed using RHB column, and the NH₃-adsorbed RHB collected from column is recycled through the soil to support plant growth.

Graphical abstract



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Introduction

Ammonia nitrogen (NH₃-N) includes both ionized ammonium (NH₄⁺) and unionized ammonia (NH₃). The global annual generation of NH₃/NH₄⁺ is estimated to exceed 200 Mt. (Khabzina and Farrusseng, 2018). A major anthropogenic source of NH₃ emissions is volatilization from agricultural systems, accounting for 56% of total global NH₃ emissions. Of this, 10–14% originates from fertilizer-derived N and 40% is attributed to N excreted by animals (Pan et al., 2016). NH₃ is produced from animal manure when urea in urine is hydrolyzed by the fecal enzyme urease, forming ammonium carbonate in an aqueous medium, which further decomposes to release another molecule of ammonia gas and carbon dioxide (Alizadeh et al., 2017). The rate of NH₃ volatilization increases with rising temperatures or elevated NH₄⁺ concentrations (Feng et al., 2022). Livestock production is the dominant source of atmospheric NH₃ emissions, accounting for around 70–80% of the total emissions (Wyer et al., 2022; EEA, 2024). Of these, a majority of this emission (50–63%) originates from animal housing, manure storage, and manure application. Rest of NH₃ emission is derived from livestock management activities such as grazing, hard

standings, feed handling, and enteric processes (EMEP/EEA, 2019; Misselbrook et al., 2023). Overall, emissions from pig houses are significant sources, accounting for 46–55% of total emissions in some countries (Gac et al., 2007; Rzeźnik and Mielcarek, 2016; Kriz et al., 2021). NH_3 emissions vary by livestock species. Estimated daily NH_3 release from housing has been estimated to be 34, 79, and 146 g NH_3 per livestock unit (LU, equivalent to 500 kg live weight) for dairy/beef cattle, fattening pigs, and laying hens, respectively (EEA, 2024).

High NH_3 concentrations in pig houses have been shown to adversely impact immunity and production performance (Philippe et al., 2011). The NH_3 excreted from pig houses emits an offensive odor and may also pose respiratory health risks to pigs, farm workers, and nearby residents (Philippe et al., 2011). Furthermore, NH_3 emissions have broader implications for ecosystem health as an atmospheric pollutant. Once deposited from the atmosphere, NH_3 contributes to eutrophication in water bodies and soil acidification due to the release of H^+ ions during nitrification (Liu et al., 2013). Additionally, $\text{NH}_3\text{-N}$ from animal manure transforms into nitrous oxide (N_2O), a greenhouse gas, via nitrification and denitrification. The current Intergovernmental Panel on Climate Change (IPCC) default methodology estimates that 1% of redeposited NH_3 is re-emitted as N_2O (IPCC, 2022). Moreover, atmospheric NH_3 enhances the atmospheric oxidizing capacity by promoting nitrous acid (HONO) generation, which leads to the formation of NH_4^+ -containing particulate matter and aerosols (Ge et al., 2019). These particulates contribute significantly to the total mass of $\text{PM}_{2.5}$ and PM_{10} (Werner et al., 2014).

For decades, various agricultural practices and techniques have been implemented to mitigate NH_3 emissions. For instance, NH_3 volatilization can be reduced through the use of acidic amendments, which maintain N in the non-volatile NH_4^+ form (Park et al., 2018), as well as microbial and enzyme inhibitors that delay the conversion of urea to NH_4^+ (Swelum et al., 2021), and the use of $\text{NH}_3\text{-N}$ adsorbing agents. Among these methods, adsorption is considered a very promising technique for capturing and removing $\text{NH}_3\text{-N}$ (Han et al., 2021). Numerous studies have demonstrated that zeolite minerals, an adsorbent material, efficiently adsorb $\text{NH}_3/\text{NH}_4^+$ (Muscarella et al., 2021; Tokushige and Ryu, 2023), thereby reducing NH_3 volatilization (Park et al., 2024a). In recent years, biochar has been widely used for ammonia-N removal in agricultural applications and wastewater treatment (Asadi et al., 2021; Han et al., 2021). Given that the recycling of underutilized feedstock may be an effective approach to achieving sustainable agriculture and environmental management, rice husk could serve as a valuable resource for biochar production. The majority of produced rice husk remains underutilized or unused (Asadi et al., 2021).

In agricultural operations, ammonia-N may be applied directly as an anhydrous N fertilizer. However, the bioavailability of NH_3 trapped in adsorbents as a fertilizing agent has seldom been estimated (Taghizadeh-Toosi et al., 2012). In this study, we hypothesized that rice husk biochar (RHB) has NH_3 adsorption capacity and that RHB-adsorbed N can be recycled through the soil matrix to support plant growth. To test the hypothesis, we performed three sequential experiments to determine: 1) the NH_3 adsorption capacity of RHB, 2) the efficiency of NH_3 removal from pig house emissions using RHB, with or without H_2SO_4 treatment, and 3) the fertilizing potential of the NH_3 -adsorbed RHB for maize growth and yield.

NH_3 -adsorbent materials

The NH_3 adsorbent used in this study was rice husk biochar (RHB) obtained from Yougi Ind Co. (Gochang, Korea), with chemical properties presented in Table 1. The RHB was produced through a controlled carbonization process conducted at 500 °C for 4 h. For the treatment of RHB with H_2SO_4 , RHB was thoroughly mixed with H_2SO_4 at a 1:2 ratio ($\text{H}_2\text{SO}_4/\text{RHB}$, v/w) in an acid-resistant container, then uniformly stirred, ensuring that no leakage or direct release of the acid solution occur. The columns

Effect of gas flow using a column

The experiments described in Section 2.2.1 were conducted to evaluate the effects of gas flow conditions (passive vs. active flow), flow rate (0.5 to 4.0 L min⁻¹), and H_2SO_4 treatment on the NH_3 adsorption capacity of RHB. In preliminary tests, wetted RHB has been found to reduce the adsorption rate and capacity of atmospheric NH_3 , likely due to the blocking of internal channels by water (Seredych et al., 2016). Based on this observation, 10 g of dry RHB (particle size: 0.5 mm–2 mm) was used to

CRediT authorship contribution statement

Sang-Hyun Park: Writing – original draft, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation. **Tae-Hwan Kim:** Writing – review & editing, Resources, Investigation, Conceptualization. **Huimang Song:** Methodology, Data curation. **Bok-Rye Lee:** Writing – review & editing, Visualization, Supervision, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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